

Time Series Analysis Final Project

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Abstract

Task 1 analyzes local and global temperatures to extract trends and seasonal patterns. Task 2 performs regime detection and trend estimation using HMMs and DLMs. Task 3 fits multivariate DLMs to deseasonalized San Francisco daily temperatures to provide short term predictions.

Task 1: Data Acquisition and Exploration

GHCN Analysis

We start by performing a time series analysis of local climate dynamics using daily temperature records from the **GHCN** archive. Our focus is on the San Francisco Downtown station (USW00023272), for which we analyze both minimum (TMIN) and maximum (TMAX) temperatures from 1921 to 2024.

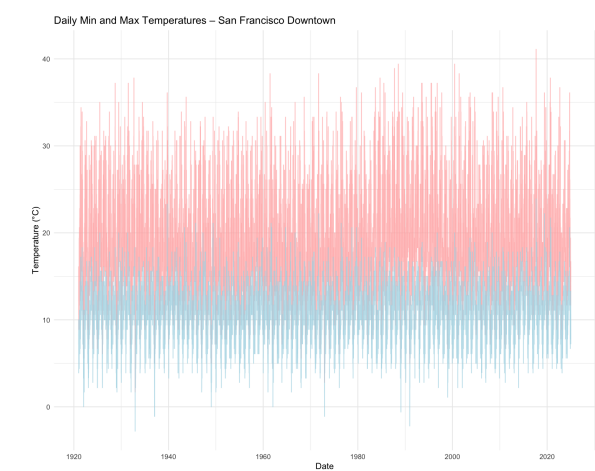


Figure 1. TMIN and TIMAX (1921-2025)

To extract trends and seasonal patterns we apply **STL decomposition** using **AIC-optimized seasonal windows**. We search for over 200 candidate window lengths and the special case "periodic" to minimize the STL residual variance (see Figure 2). The window lengths were independently selected for the TMIN and TMAX series.

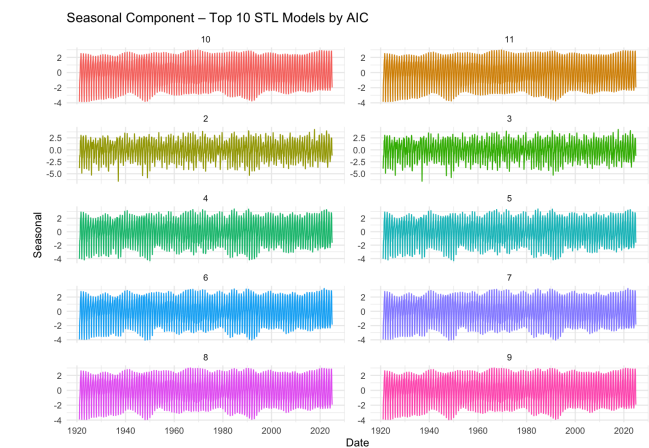


Figure 2. STL decomposition with AIC-optimized seasonal window

The resulting **decomposition** (see Figure 3) reveals a very mild warming trend in TMAX after 1970, while TMIN is characterized by a flat evolution. An analysis of the **daily temperature range** confirms the asymmetry in warming dynamics. Both series display stable, high-variance seasonal cycles, and low-variance, weakly autocorrelated remainders (note that the remainders plotted are obtained as the ratio between the remainder and the raw values of the series.).

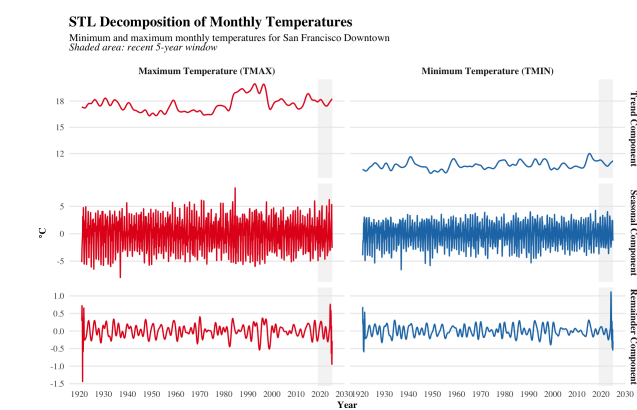


Figure 3. STL optimal window structure for each variable

To formally analyze extremes, we fit **Generalized Extreme Value (GEV) models** (see Figure 4). These return levels quantify the severity and rarity of extreme temperature events. An extreme event such as 40°C in TMAX is expected to occur on average once every one hundred years. For TMIN, a high temperature 6°C is also a 100-years event. For DTR, a diurnal range of 25°C is expected once every one hundred years, showing that large daily temperature swings are rare but possible.

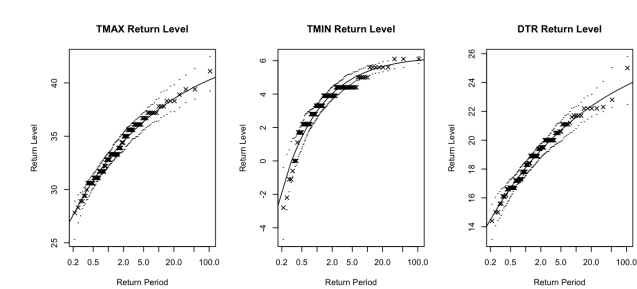


Figure 4. GEV return level plots for TMAX, TMIN, and DTR

Seasonality and Time-Frequency Dynamics.

We first plot the raw time series on a small subsample and smooth the data to get an idea of how temperatures evolve.

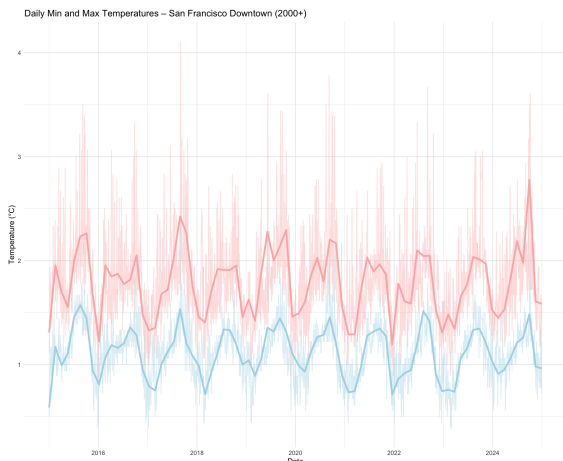


Figure 5. Restricted sample data with LOESS smoothing

We quantify persistence in seasonal dynamics through **lag-12 autocorrelations**: $ACF_{12}(TMIN) = 0.787$, $ACF_{12}(TMAX) = 0.769$. To analyse the temporal evolution of such patterns, we compute **rolling 10-year autocorrelations at lags 1–48**, highlighting lag 12 in red and color-coding other lags by distance (see Figure 6). Seasonality at lag 12 is the strongest across lags and was strong throughout the whole period.

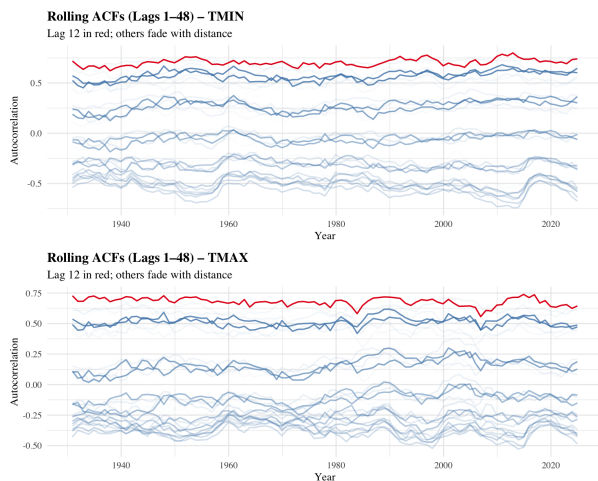


Figure 6. 12 months rolling ACF

We also apply **wavelet power spectrum analysis** to TMIN and TMAX. Both spectra reveal a dominant and persistent annual band (12-month period) across the full time span with some weaker components appearing occasionally.

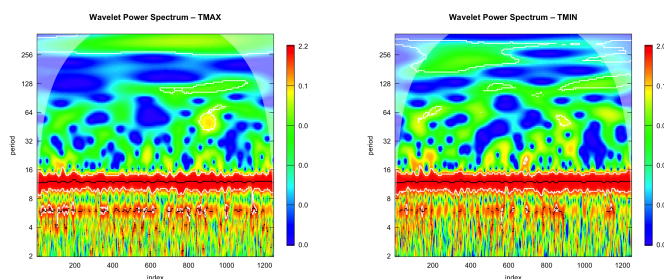


Figure 7. Wavelet TMIN

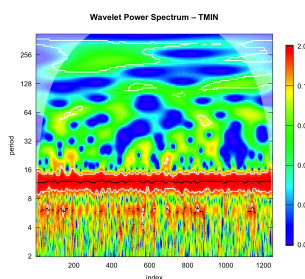


Figure 8. Wavelet TMAX

Spectral density analysis confirm the findings above by showing that both the minimum and maximum temperatures have a dominant annual cycle, with a sharp peak at frequency (that is, 1 cycle / year) (see Figure 9). Secondary peaks at integer multiples (like 2, 3 and so on) are coherent to the analysis above. The flatness elsewhere confirms low contribution from other frequencies; temperature variation is overwhelmingly driven by yearly periodicity.

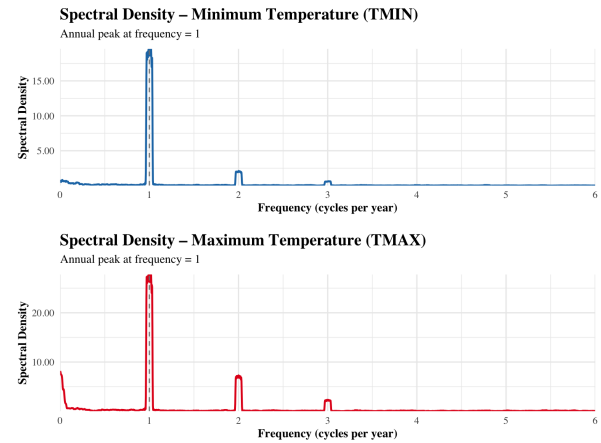


Figure 9. Wavelet spectrum of TMAX

Next, we plot **monthly values for one year every ten** (upper panel Figure 10), and compute **decadal monthly averages** as well (lower panel Figure 10). These reveal consistent seasonal cycles, with temperatures increasing in spring and summer. We also produce animated visualizations of (i) **annual seasonality** and (ii) **polar plots**, which can be found in our code.

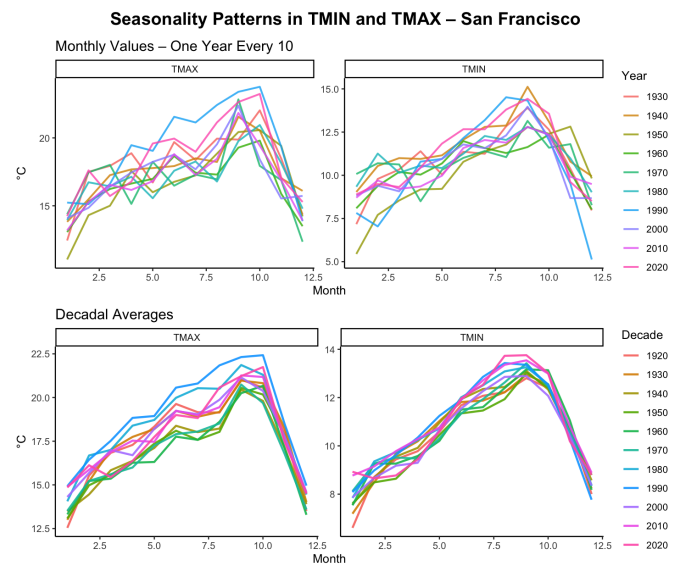


Figure 10. Seasonality

GISTEMP Analysis

We analyze long-term global temperature anomalies using the **GISTEMP** dataset, reporting monthly deviations from the 1951–1980 baseline values based on land and sea station records. The dataset spans 1880 to 2024.

We plot the kernel density estimates of monthly anomalies for each decade (Figure 11), to show how global temperatures changed over this time frame. In the early decades (1880s to 1930s), temperatures were centered around -0.5°C , while in the 2010s and 2020s they were centered above 0.8°C . In addition, there was also an increase in the variability of the temperature. These patterns highlight a move not only toward a warmer climate but also to a more volatile global climate.

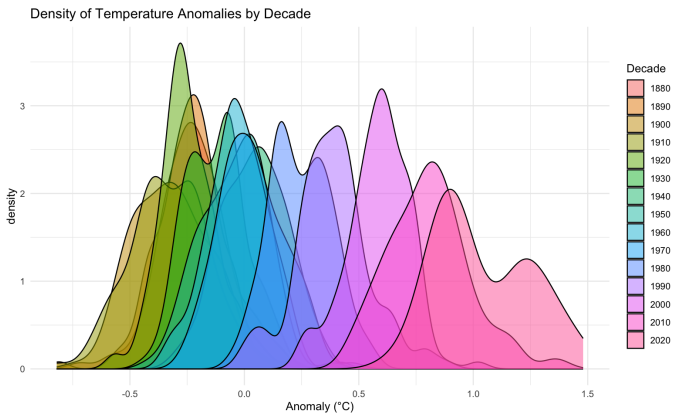


Figure 11. Density plot by decade

We begin with **STL** decomposition, evaluating seasonal windows from 1 to 200 and the “periodic” specification. The optimal configuration, **selected by minimum AIC**, sets $s.window = 3$, indicating that tight seasonal smoothing best captures the model seasonal variation. In figure 12 the trend component displays a nonlinear upward pattern beginning in the mid-20th century and sharply accelerating post-1970, consistent with anthropogenic forcing. The seasonal component is high-frequency, low-amplitude, and stable, reflecting persistent annual cycles. The residual component is stationary, low-variance, and centered, suggesting that the decomposition is well-specified and structurally sound (note that again we have divided the residual by the raw ts value).

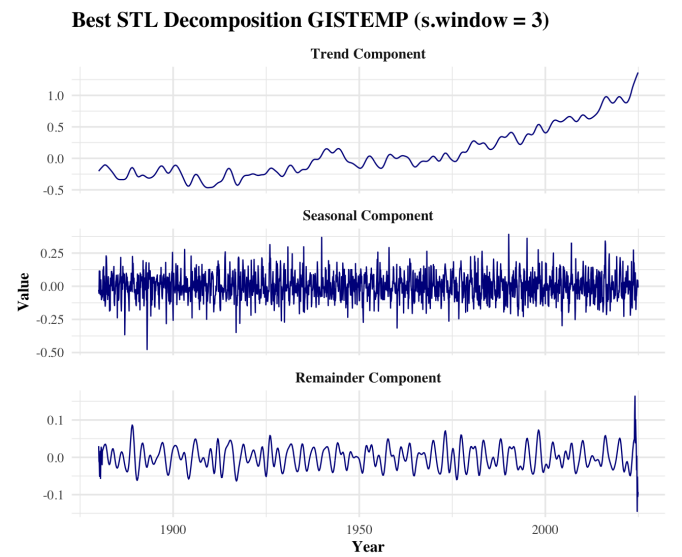


Figure 12. GISTEMP Decomposition

We apply the **Bai-Perron multiple breakpoint algorithm** to capture long run changes in the mean structure. Five

significant structural breaks around 1903, 1939, 1978, 1998, and 2015 divide the series into six regimes: flat in 1880–1903, mild cooling until 1939, weak warming until 1978, followed by strong and then accelerating warming (Figure 13).

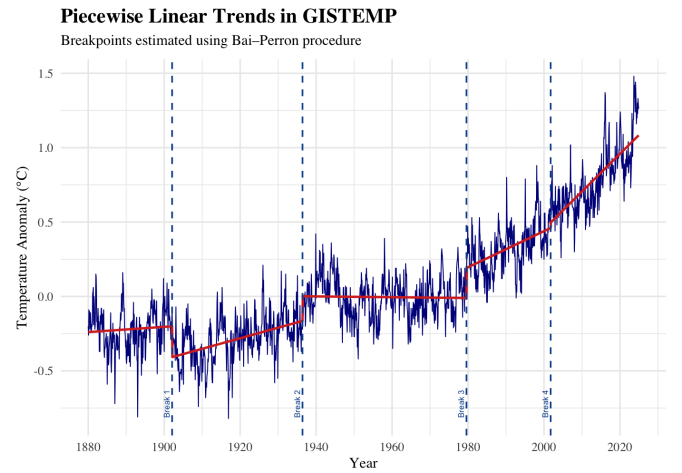


Figure 13. OLS fitted on breakpoints

For robust trend–cycle separation, we fitted and compared 3 **Hodrick–Prescott (HP)** filter variants: the standard fixed- λ filter, a breakpoint-adjusted version using **Bai–Perron** regimes, and an adaptive HP filter with time-varying λ_t . The adaptive version performs best, and it explains 75% of the total series variance through its trend component. An **Empirical Ensemble Mode Decomposition (EEMD)** confirms this result. The IMFs represent oscillatory patterns from monthly to interdecadal timescales, while the residue matches the upward convex shape seen in STL and HP filters. EEMD provides further evidence for the nonstationary, multi-scale structure of global temperature anomalies.

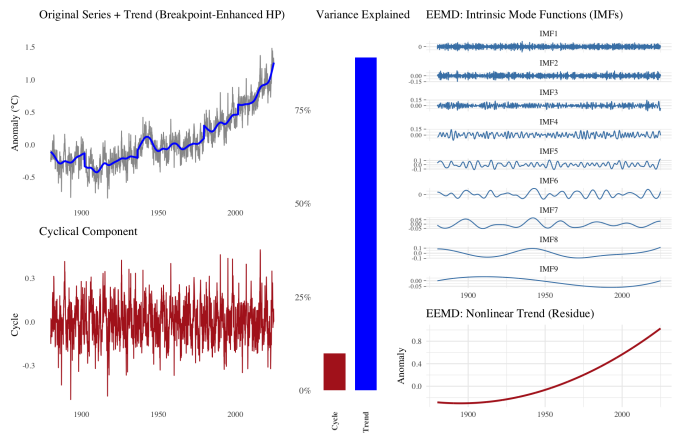


Figure 14. Trend and cycle via HP and EEMD decomposition

Seasonality and Time–Frequency Dynamics.

We produced the following analysis

- Decadal seasonal cycles:** we overlayed one year out of 10 (Figure 15) left panel) and the decadal averages (Figure 15) right panel)

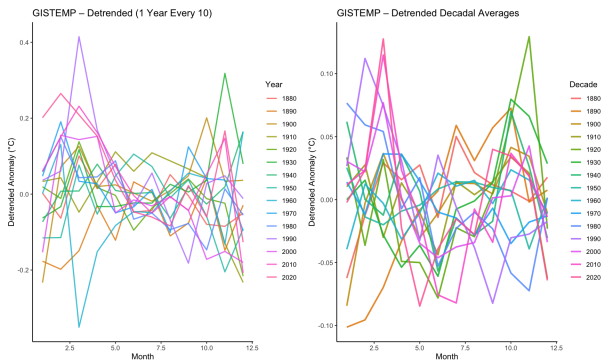
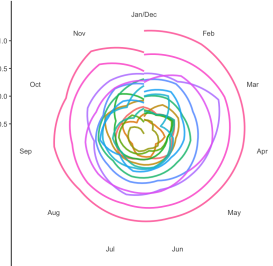


Figure 15. Decadal seasonal analysis

- **Polar plots** to visualize intra-annual cycles over time.

GISTEMP - Static Polar Plot (1 in 10 Years)



- **Animated visualizations** of both.
- **Average seasonal effect by month extracted by a complete dlm model** including level trend and seasonal effects. In such case, we find positive seasonal anomalies in february, march, october, and november, and negative anomalies in may and june.

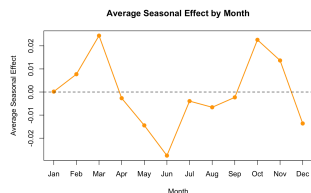


Figure 16. Average seasonal effect by month

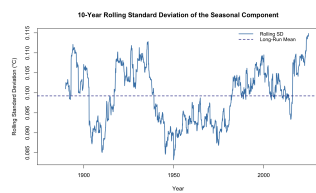


Figure 17. Rolling sd of the seasonal component

enables regime changes to be estimated endogenously from the data.

Data Preparation We use monthly global temperature anomalies from the GISTEMP dataset. Seasonality has been removed via STL decomposition, producing a deseasonalized series. Several covariates have been engineered to support richer HMM structures:

- A standardized `time_variable` to allow transition probabilities to vary over time.
- A `lag_anomaly` to capture autoregressive emissions.
- A factor `month` for seasonal emission effects.

Modeling Strategy We implement and estimate the following **seven HMM families**:

- **Constant Mean**: A baseline HMM where each state's emissions have a constant mean, and the transition matrix is fixed over time.
- **Time Varying Transition**: Transition probabilities change smoothly over time, modeled as a logistic function of time; emission means remain constant across states.
- **Autoregressive Emission**: Emissions follow a state-dependent AR(1) process, while transition probabilities remain fixed.
- **Seasonal Emission**: Emission means vary by calendar month within each state; transitions are time-invariant.
- **Fully Flexible**: Emissions depend on both the lagged anomaly and month, transition probabilities also vary over time.
- **Linear Trend Emission**: Within each state, emission means follow a linear trend over time.
- **Quadratic Trend Emission**: Emission means evolve quadratically over time within each state, allowing for acceleration or deceleration in trends.

All models are estimated for 2 to 5 states, with multiple random starting points to avoid local optima. Model selection was based on AIC and BIC.

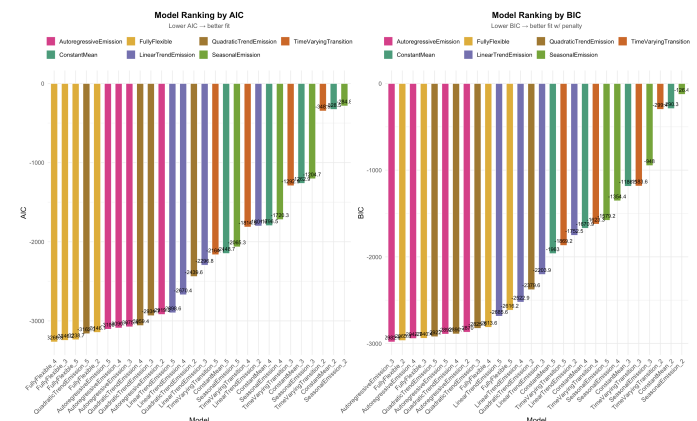


Figure 18. BIC scores for HMMs with varying structures.

Model Comparison and Selection According to BIC (Figure 18), the best-performing model is the **Autoregressive Emission (3 States)** model, followed closely by the **Fully Flexible (2 States)** model. For the top model, we decode the most likely sequence of hidden states (Figure 19).

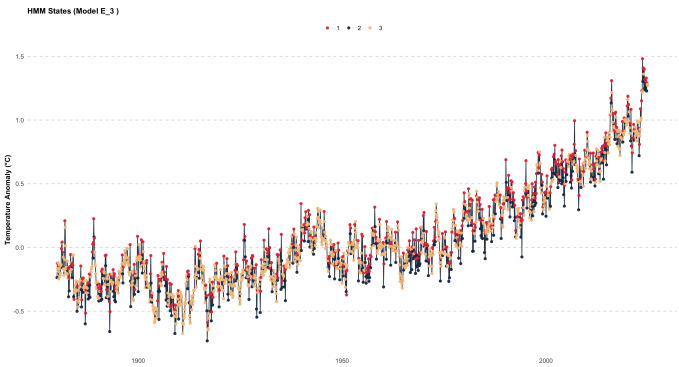


Figure 19. Posterior decoding for selected model.

Transition Matrix and Time-Varying Transitions For such analysis, we resort to the constant mean and the time varying transition models. For the **constant mean (5 states)** model, we extract the fixed transition matrix. High diagonal elements confirm the persistence of each regime (Figure 20).

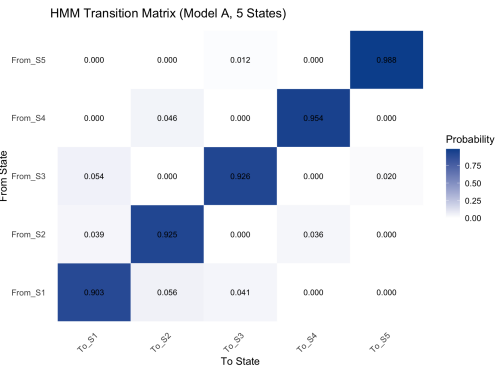


Figure 20. Estimated transition matrix (Model: constant_mean, 5 states).

For the **time varying transition (5 states)** model, we compute and visualize transition probabilities as functions of time. Each row of the transition matrix varies over time, revealing increasing lock-in of warm regimes (Figure 21). We also **animated the full transition matrix**.

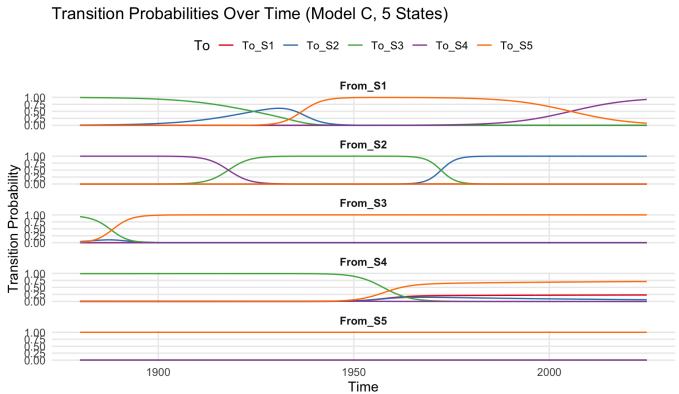


Figure 21. Time-varying transition probabilities transition matrix

Extended Analysis: Regime Features (Model: time_varying_transition) We have analyzed key characteristics of the latent states (Figure ??):

- **State durations:** number of months the system stays in each regime.
- **Mean anomaly:** average temperature anomaly by state.
- **Volatility:** standard deviation of anomalies in each state.
- **Transition frequencies:** frequency of switching from one regime to another.

Recent regimes (in red, Figure 22) display high anomalies and higher volatility of the anomaly within the state, consistent with sustained warming.

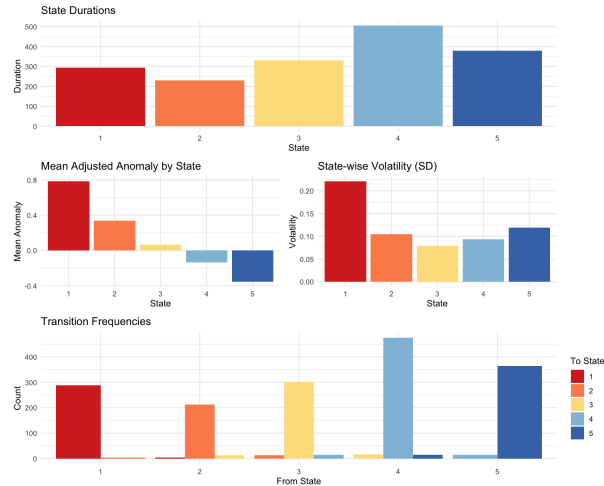


Figure 22. Regime analysis: durations, means, volatility, and transitions (Model: time_varying_transition, 5 states).

Model Comparison: Constant Mean vs. Linear Trend Emissions Figure 23 contrasts two (easily interpretable) competing HMM specifications:

- **Constant Mean (3 States):** Each regime has a fixed mean anomaly, and transitions between discrete levels explain changes in the series.
- **Linear Trend Emission (2 States):** Each regime allows a linear time trend within-state, capturing gradual warming patterns.

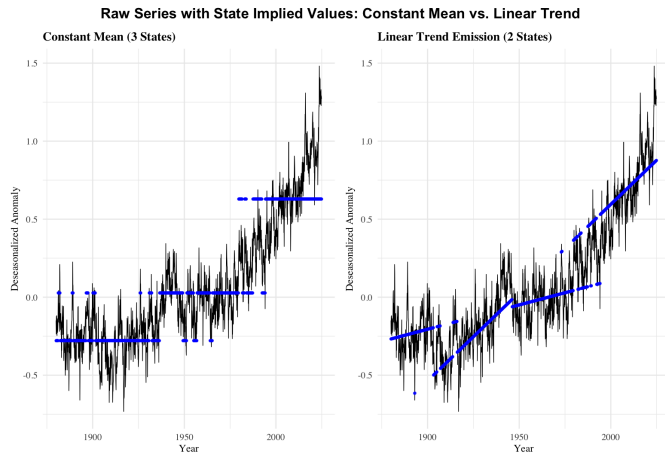


Figure 23. Raw Series with State-Implied Values: Constant Mean vs. Linear Trend Emissions

The left panel shows that the **Constant Mean** model struggles to explain the long-term structure of the anomaly series.

Although it captures some level shifts (e.g., around 1980), it approximates the underlying trend as a sequence of abrupt jumps, leading to a coarse and unrealistic segmentation.

By contrast, the right panel shows that the **Linear Trend Emission** model explains the gradual rise in anomalies more faithfully. The inclusion of a within-state slope allows the model to describe persistent trends without requiring excessive state changes.

This comparison highlights a key insight: **when structural change is gradual rather than abrupt**, using trend-based emission models is crucial. The Constant Mean HMM misrepresents the warming trajectory as regime shifts, while the Trend model provides a smoother and more realistic decomposition of global warming dynamics.

Final Evaluation of HMM approach

Strengths:

- Flexibly detects regime shifts or trend breaks.
- Allows both abrupt jumps and smooth transition dynamics.
- Generates interpretable posterior state paths.

Limitations:

- Gaussian assumptions may under-represent extreme values.
- Sharp regime changes may oversimplify gradual warming trends.
- Complex models (e.g., fully flexible) risk overfitting and interpretability loss.

Conclusion Hidden Markov Models applied to deseasonalized GISTEMP data provide strong evidence of climate regime change. Several models, especially **time varying transition (5 states)**, uncover increasing persistence in high-temperature regimes since the 1980s. This structural evolution aligns with anthropogenic global warming, and HMMs allow both sharp and smooth change detection in a unified probabilistic framework.

Step 2. Dynamic Linear Models

To analyze the presence of long-term acceleration or deceleration in global warming, we estimate two core **Dynamic Linear Models (DLMs)**. First, we fit a **Random Walk Plus Noise (RWPN) model**, which assumes that the underlying level of the temperature anomaly follows a random walk, allowing for smooth shifts in the mean without explicitly modeling a trend component. Second, we estimate a **Locally Linear Trend (LLT) model**, which extends the RWPN structure by incorporating both a level and a slope component. This specification allows us to directly capture not only the average level of global temperature anomalies, but also its time-varying rate of change. Both models are estimated by maximum likelihood using and filtered/smoothed using Kalman techniques. In addition to trend models, we estimate two specifications that explicitly model seasonal structure. First, the **Classical Seasonal DLM (d1mModSeas)**: the strong in-sample fit (lowest RMSE) reflects its ability to tightly track recurring monthly patterns, but the model to high complexity and is prone to instability. Second, the **Fourier Seasonal DLM**: this specification captures seasonality using sine and cosine terms (i.e.,

trigonometric basis functions). Unlike the classical model, it assumes the seasonal effect is smooth and cyclical. This leads to greater parsimony and potentially better out-of-sample behavior. However, the Fourier model may underfit if real-world seasonality deviates from purely sinusoidal forms. Note that the first two models are fitted on deseasonalized data while the third and the fourth are fit on the raw data.

Table 1. Model comparison: likelihood, information criteria, and RMSE.

Model	Log-Likelihood	AIC	BIC	RMSE
RWPN (Random Walk Plus Noise)	3188.6	-6373	-6362	0.0544
LLT (Locally Linear Trend)	3135.9	-6266	-6249	0.0547
Classical Seasonal DLM	-541.3	1091	1113	0.0100
Fourier Seasonal DLM	2799.5	-5591	-5569	0.0657

Model selection discussion The RWPN model yields the best fit in terms of AIC/BIC and slightly lower RMSE than the LLT. Seasonal DLMs (especially classical) exhibit much lower RMSE due to their strong periodic structure, but fit the overall dynamics less effectively, as shown by poor likelihood scores (Table 1). This highlights the trade-off between short-term fit and capturing structural long-term trends.

In all the four models, we extract and visualize the smoothed states using Kalman smoothing (Figure 24). All models succeeded in detecting an underlying increasing trend in temperatures.

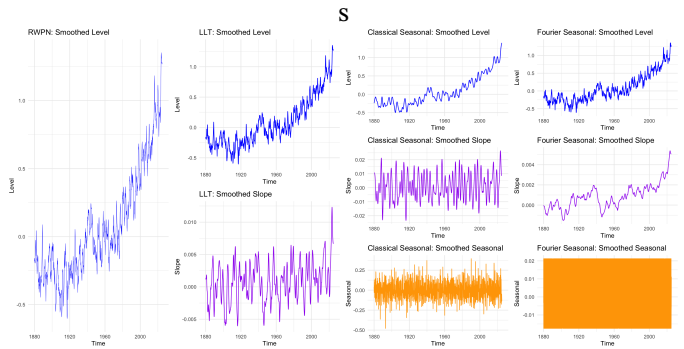


Figure 24. Smoothed level and slope from RWPN and LLT models.

The figure below shows the trend estimated through the classical seasonal dlm overlaid on the raw time series.

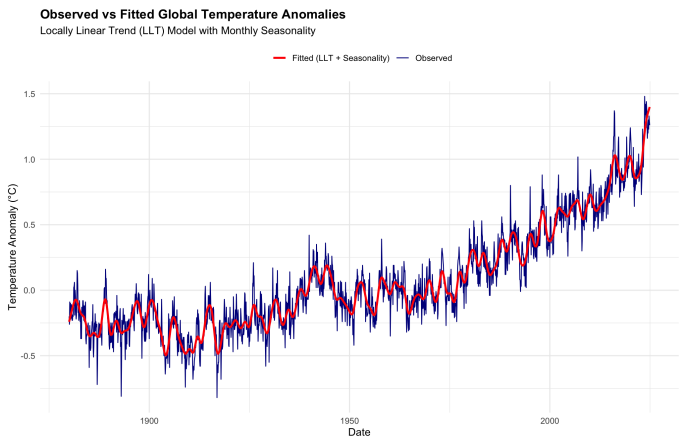


Figure 25. Trend and confidence intervals from DLM smoothing.

Remark, **Variance decomposition** from the seasonal DLM:

- The **trend component** (level + slope) explains **93.77%** of the total variation.
- The **seasonal component** accounts for only **6.02%**.

This confirms that long-run structural changes, rather than seasonality, dominate the evolution of global temperature anomalies.

DLMs vs HMMs To evaluate the performance of these DLMs, we compare them with the HMM approach from Step 1. HMMs segment the time series into discrete regimes, each with its own mean anomaly, and are well-suited for capturing abrupt shifts (e.g., transitions between cooling and warming). In contrast, DLMs offer a continuous-time representation that captures gradual structural evolution, including changes in the rate of warming.

While HMMs highlight sharp breakpoints, DLMs, particularly the LLT model, provide interpretable parameters that reveal the smooth trend evolution of the series.

In summary, HMMs and DLMs are complementary: the former detects discrete shifts, while the latter models gradual, long-term changes. Used together, they offer a more nuanced view of global temperature dynamics.

Task 3. Weather Forecasting

Objective. The daily maximum and minimum temperature data starting from the first of January 1921 were taken from the San Francisco Downtown station in GHCN database. After deseasonalizing the series, three different **Dynamic Linear Model (DLM)** were applied.

Modeling Strategy. We estimate three increasingly structured DLMs:

- **Model A: Independent RWPN.** Random Walk plus Noise is the simplest DLM model for univariate time series and we apply it by assuming that the maximum and minimum temperatures evolve independently with no interaction. Moreover, **RWPN are appropriate for deseasonalized series which show no clear trend.**
- **Model B: Seemingly unrelated RWPN.** It is an extension of Model A. The two series are no longer considered univariate, but rather multivariate time series with underlying state noise correlated. The levels of the time series are correlated while the observation noises are still treated as independent.
- **Model C: Latent Factor RWPN.** It introduces a **shared latent factor** ξ_t influencing both the maximum and minimum temperature time series. A loading matrix F links the latent factor to observed temperatures, offering a parsimonious and flexible representation.

Parameter Estimates.

Since the system matrices are unknown for each of the three models, we estimate the parameters through the Maximum Likelihood framework. The estimated matrices are then used as if they were known constant in their respective models.

Table 2. Estimated parameters for each model.

Model	State-Noise Variances	Observation-Noise Variances	Other Parameters
(a) Independent RWPN	$\sigma_{w,1}^2 = 0.002098$ $\sigma_{w,2}^2 = 0.023647$	$\sigma_{v,1}^2 = 0.002370$ $\sigma_{v,2}^2 = 0.002599$	—
(b) SUR RWPN	$\sigma_{w,11}^2 = 0.002137$ $\sigma_{w,22}^2 = 0.020984$ $\rho_w = 0.8695$	$\sigma_{v,1}^2 = 0.002469$ $\sigma_{v,2}^2 = 0.004187$	—
(c) Latent Factor RWPN	$\sigma_w^2 = 0.022950$	$\sigma_{v,1}^2 = 0.157351$ $\sigma_{v,2}^2 = 0.002994$	$\mu_1 = 9.3265$ $\mu_2 = 18.0712$ $\beta = 0.3373$

Forecast Evaluation All models are smoothed using the Kalman filter. One-step-ahead forecasts are evaluated on a hold-out test set using RMSE and MAE.

Table 3. Forecast performance (deseasonalized temperatures).

Model	RMSE (TMIN)	RMSE (TMAX)	MAE (TMIN)	MAE (TMAX)
Independent RWPN	0.7034	0.9518	0.5951	0.7080
SUR RWPN	0.7039	0.9516	0.5958	0.7078
Latent Factor RWPN	0.5025	0.9532	0.3494	0.7095

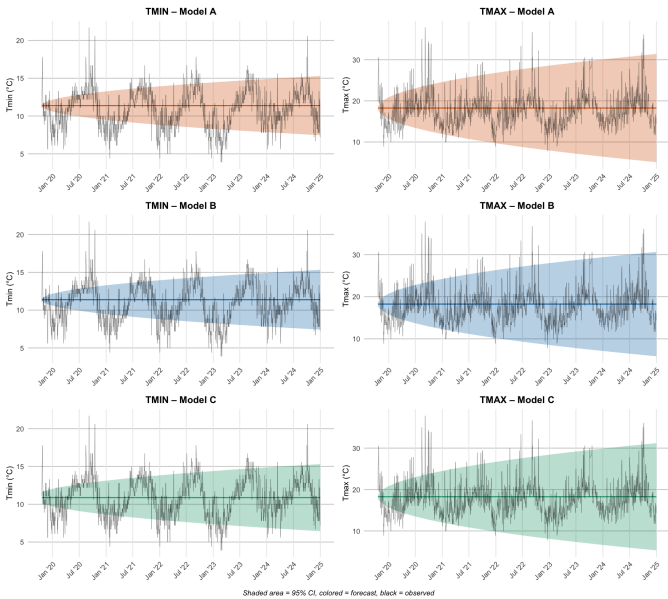


Figure 26. Forecast On the Test set

Interpretation Despite the near-identical forecast performance across models, their structural assumptions provide different insights. The SU RWPN model reveals a strong correlation in latent innovations ($\rho_w \approx 0.87$), suggesting that shared meteorological drivers impact both series. The latent factor model embeds this commonality into a single stochastic trend, weighted differently across TMIN and TMAX. The loading estimate $\beta \approx 0.30$ shows that minimum temperature is only partially influenced by the shared factor, whereas TMAX reflects it more directly. This supports the interpretation of a dominant underlying atmospheric condition—such as cloud cover or regional pressure patterns—governing both variables, with stronger effects on daytime extremes.

Extra: Reconstruction of Raw-Scale Forecasts Using Training-Only Seasonality and prediction.

After computing one-step-ahead forecasts on the deseasonalized scale, we **re-add seasonal components estimated during the training phase** through our `stl 3`.

- For each test-set observation (with known day-of-year d), we use a precomputed lookup table from the training data to retrieve average seasonal values $S_{\min,d}$ and $S_{\max,d}$.
- These are added to the predicted values $\hat{X}_{t|t-1}$ to form raw-scale forecasts:

$$\hat{T}_{\min,t} = \hat{X}_{\min,t} + S_{\min,d}, \quad \hat{T}_{\max,t} = \hat{X}_{\max,t} + S_{\max,d}$$

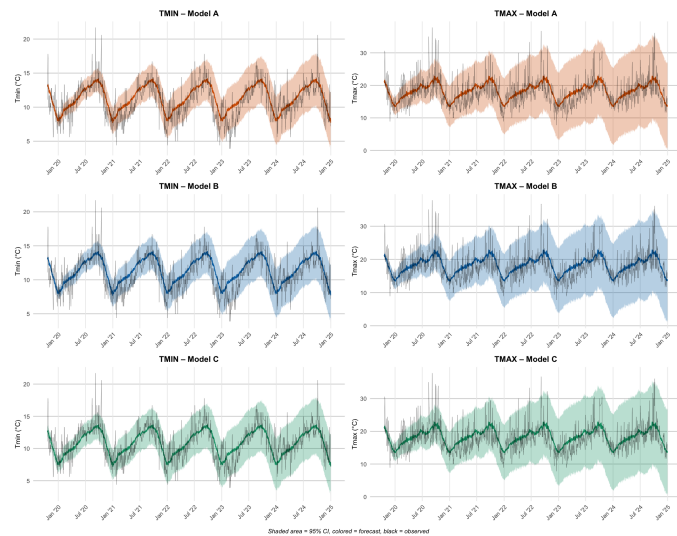


Figure 27. Forecast On the Test set Seasonality Expanded

Finally, we plot our prediction for the end of 2025 - beginning of 2026.

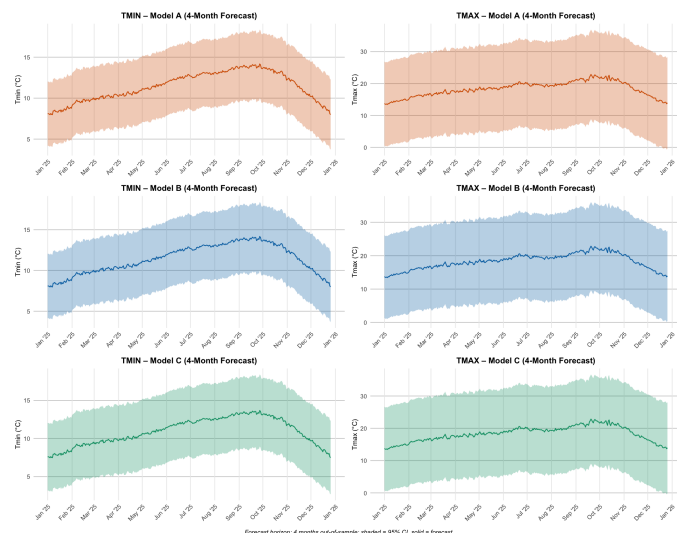


Figure 28. Forecast on 2026

Appendix: Spatio-Temporal Temperature Modeling with GHCN Data

Preprocessing and Initial Visualization. We begin by aggregating the raw GHCN daily data to monthly frequency for

initial exploration. Monthly averages of TMIN and TMAX are computed using floor-rounded month indicators and divided by 10 to convert from tenths of degrees Celsius. To facilitate visualization and computation, we restrict our sample to January 2010 onwards. Station coordinates are extracted and plotted over a U.S. map to assess spatial coverage and identify potential clustering (Figure 29).

Stations on U.S. Map (Daily Data, Colored by Residual)

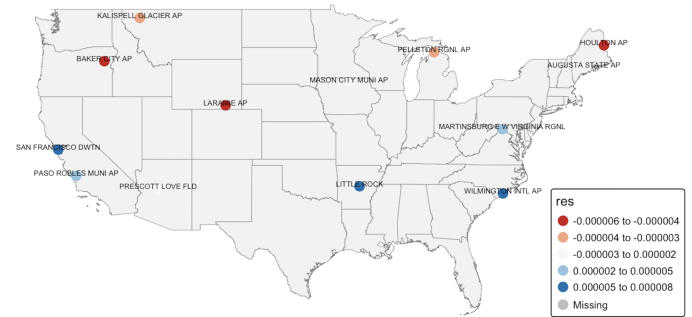


Figure 29. Stations across the U.S.

We then reshape the dataset into wide matrices for TMIN and TMAX, with months as rows and stations as columns. First, we analyze spatio-temporal patterns allowing us to see how temperature evolved across time and across multiple stations (see, for example, Figure 30 for TMIN)

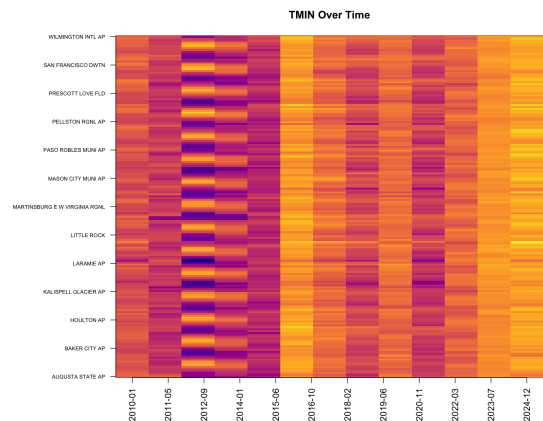


Figure 30. T-MIN across Space and Time

To quantify spatial dependencies, we compute correlation matrices across stations for both TMIN and TMAX (see for example Figure 31). Heatmaps revealed strong positive correlation among geographically proximate stations, justifying spatial smoothing in modeling.

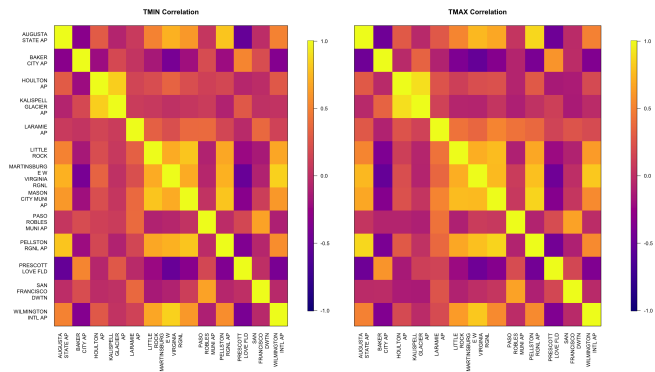


Figure 31. Cross Station correlations

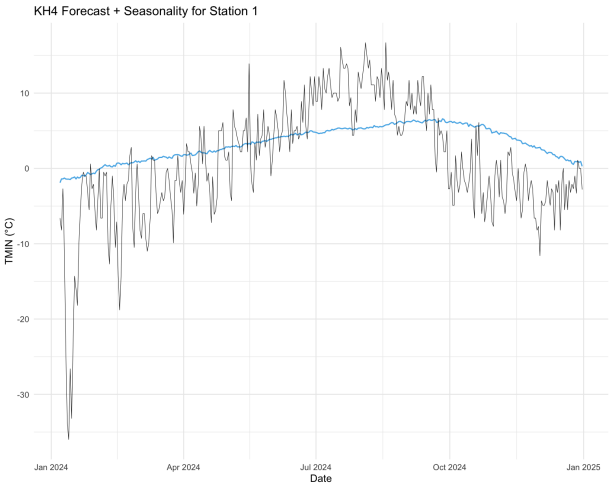


Figure 32. Forecast on Test set

Model Specification and Estimation. For full-resolution modeling, we return to the daily data, assigning each date a unique index (`id.date`), each station a spatial index (`id.zone.int`), and their interaction a unique spatio-temporal identifier (`id.zoneu.dateu`). We use four models for our analysis: KH1, KH2, KH3, and KH4, each increasing in space–time complexity. **KH1** includes: (i) a global spatial effect (BYM), (ii) a smooth temporal trend (RW1), and (iii) independent noise in space and time (IID). **KH2** extends this by allowing a separate spatial pattern for each month ($\text{Besag} \times \text{month}$). **KH3** instead introduces station-specific temporal dynamics ($\text{RW1} \times \text{station}$). **KH4** combines both extensions. It includes: (i) a structured spatio-temporal interaction ($\text{Besag} \times \text{RW1}$), (ii) a global time trend (RW1), (iii) unstructured temporal noise (IID), and (iv) unstructured spatial noise (IID). Posterior summaries of fixed effects and precision parameters are extracted and are reported below.

Table 4. Model selection criteria for KH1–KH4 (lower is better)

Model	DIC	WAIC
KH1	412370.2	412406.6
KH2	412384.0	412410.2
KH3	412124.3	412146.5
KH4	412516.5	412558.8

Residuals from kh4 (observed minus predicted TMIN) are visualized using `tmap` to assess spatial prediction performance (Figure 29 above). Residuals are computed as the difference between observed and fitted daily TMIN values at each station, then averaged over time. It is interesting to see how forecast errors seems to be connected to latitude.

Forecasting and Model Interpretation. Finally, we forecast on a held out test set with the kh4 model, and in figure we add back the seasonality extracted on the train set through `stl 3` (Figure 32).